

Test Cases

SauceDemo E-commerce Platform

Project: SauceDemo Test Automation

Document Owner: QA Engineer

Total Test Cases: 52

1. Document Overview

This document contains detailed test cases for SauceDemo e-commerce website. Test cases are organized by function areas and connected to user stories. Each test case includes detailed steps, expected results, and connection to requirements.

2. Test Case Template

Test Case Structure:

- **Test Case ID:** Unique number
- **Test Case Title:** Clear name
- **User Story:** Connection to user story
- **Priority:** Critical/High/Medium/Low
- **Severity:** Critical/High/Medium/Low
- **Test Type:** Functional/Non-functional/Integration
- **Preconditions:** What needs to be ready before test
- **Test Steps:** Detailed steps to follow
- **Expected Result:** What should happen
- **Post-conditions:** What happens after test

3. Test Cases by Function Area

3.1 Authentication and Login (12 Test Cases)

TC-001: Successful Login with Valid User

- **User Story:** US-1.1.1
- **Priority:** Critical
- **Severity:** High
- **Test Type:** Functional
- **Preconditions:**
 - SauceDemo website is accessible
 - Valid user credentials are available (standard_user/secret_sauce)
- **Test Steps:**
 - Open <https://www.saucedemo.com>
 - Enter valid username "standard_user"
 - Enter valid password "secret_sauce"
 - Click "Login" button
- **Expected Result:**
 - User is moved to inventory page
 - URL changes to "/inventory.html"
 - Product catalog is displayed
 - Shopping cart icon is visible
- **Post-conditions:** User is logged in and session is active

TC-002: Login with Invalid Username

- **User Story:** US-1.1.2
- **Priority:** High
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** SauceDemo website is accessible
- **Test Steps:**
 - Open <https://www.saucedemo.com>
 - Enter invalid username "invalid_user"
 - Enter valid password "secret_sauce"
 - Click "Login" button
- **Expected Result:**
 - Error message appears: "Epic sadface: Username and password do not match any user in this service"
 - User stays on login page
 - Login form can be used again

- **Post-conditions:** User is not logged in

TC-003: Login with Invalid Password

- **User Story:** US-1.1.2
- **Priority:** High
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** SauceDemo website is accessible
- **Test Steps:**
 - Open <https://www.saucedemo.com>
 - Enter valid username "standard_user"
 - Enter invalid password "invalid_password"
 - Click "Login" button
- **Expected Result:**
 - Error message appears: "Epic sadface: Username and password do not match any user in this service"
 - User stays on login page
 - Username field still contains entered value
- **Post-conditions:** User is not logged in

TC-004: Login with Locked Out User

- **User Story:** US-1.1.3
- **Priority:** High
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** SauceDemo website is accessible
- **Test Steps:**
 - Open <https://www.saucedemo.com>
 - Enter locked username "locked_out_user"
 - Enter valid password "secret_sauce"
 - Click "Login" button
- **Expected Result:**
 - Error message appears: "Epic sadface: Sorry, this user has been locked out."
 - User stays on login page
 - Access is denied
- **Post-conditions:** User is not logged in

TC-005: Login with Empty Username

- **User Story:** US-1.1.4
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** SauceDemo website is accessible
- **Test Steps:**
 - Open <https://www.saucedemo.com>
 - Leave username field empty
 - Enter valid password "secret_sauce"
 - Click "Login" button
- **Expected Result:**
 - Error message appears: "Epic sadface: Username is required"
 - User stays on login page
 - Password field is cleared
- **Post-conditions:** User is not logged in

TC-006: Login with Empty Password

- **User Story:** US-1.1.4
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** SauceDemo website is accessible
- **Test Steps:**
 - Open <https://www.saucedemo.com>
 - Enter valid username "standard_user"
 - Leave password field empty
 - Click "Login" button
- **Expected Result:**
 - Error message appears: "Epic sadface: Password is required"
 - User stays on login page
 - Username field still contains entered value
- **Post-conditions:** User is not logged in

TC-007: Login with Both Fields Empty

- **User Story:** US-1.1.4
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** SauceDemo website is accessible
- **Test Steps:**
 - Open <https://www.saucedemo.com>
 - Leave both username and password fields empty
 - Click "Login" button
- **Expected Result:**
 - Error message appears: "Epic sadface: Username is required"
 - User stays on login page
 - Both fields stay empty
- **Post-conditions:** User is not logged in

TC-008: Successful Logout

- **User Story:** US-1.2.2
- **Priority:** High
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is logged in and on inventory page
- **Test Steps:**
 - Click hamburger menu (three lines icon)
 - Click "Logout" from menu
- **Expected Result:**
 - User is moved back to login page
 - Session is ended
 - URL changes to base URL
 - Login form is displayed
 - **Post-conditions:** User is logged out and stayed on login page

TC-009: Session Stays Active During Navigation

- **User Story:** US-1.2.1
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is logged in
- **Test Steps:**
 - Go to inventory page
 - Click on a product to view details
 - Go back to inventory
 - Add item to cart
 - Go to cart page
 - Return to inventory
- **Expected Result:**
 - Session stays active during all navigation
 - User stays logged in
 - Cart contents are saved
 - No re-login is needed
- **Post-conditions:** User session is maintained

TC-010: Login with Problem User

- **User Story:** US-1.1.1
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** SauceDemo website is accessible
- **Test Steps:**
 - Open <https://www.saucedemo.com>
 - Enter username "problem_user"
 - Enter password "secret_sauce"
 - Click "Login" button
- **Expected Result:**
 - User is moved to inventory page
 - Login is successful
 - Product catalog is displayed (may have incorrect images)

- **Post-conditions:** Problem user is logged in

TC-011: Login with Performance User

- **User Story:** US-1.1.1
- **Priority:** Medium
- **Severity:** Low
- **Test Type:** Performance
- **Preconditions:** SauceDemo website is accessible
- **Test Steps:**
 - Open <https://www.saucedemo.com>
 - Enter username "performance_glitch_user"
 - Enter password "secret_sauce"
 - Click "Login" button
 - Measure response time
- **Expected Result:**
 - User is moved to inventory page
 - Login is successful but may be slower
 - Response time is measured and recorded
- **Post-conditions:** Performance user is logged in

TC-012: Error Message Can Be Dismissed

- **User Story:** US-1.1.2
- **Priority:** Low
- **Severity:** Low
- **Test Type:** Functional
- **Preconditions:** Error message is shown on login page
- **Test Steps:**
 - Create login error (use wrong credentials)
 - Check that error message is displayed
 - Click "X" button on error message
- **Expected Result:**
 - Error message disappears
 - Login form can be used again
 - User can try login again
- **Post-conditions:** Error message is cleared

3.2 Product Catalog Management (10 Test Cases)

TC-013: Display Product Inventory

- **User Story:** US-2.1.1
- **Priority:** Critical
- **Severity:** High
- **Test Type:** Functional
- **Preconditions:** User is logged in and on inventory page
- **Test Steps:**
 - Check that inventory page is displayed
 - Count total number of products
 - Check that each product shows image, name, description, and price
- **Expected Result:**
 - 6 products are displayed
 - Each product has image, name, description, and price
 - Products are arranged in grid layout
 - All product information is visible
- **Post-conditions:** Product inventory is fully displayed

TC-014: Product Detail View

- **User Story:** US-2.1.2
- **Priority:** High
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is logged in and on inventory page
- **Test Steps:**
 - Click on "Sauce Labs Backpack" product name
 - Check that product detail page is displayed
 - Check that product information is complete
 - Check that "Add to cart" button is present
- **Expected Result:**
 - Product detail page is displayed
 - Product image is larger and clear
 - Product name, description, and price match inventory page
 - "Add to cart" button is visible and works
 - "Back to products" link is available

- **Post-conditions:** Product detail page is accessible

TC-015: Sort Products by Name (A to Z)

- **User Story:** US-2.1.3
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is logged in and on inventory page
- **Test Steps:**
 - Click on sort dropdown
 - Select "Name (A to Z)"
 - Check product order
- **Expected Result:**
 - Products are sorted alphabetically by name
 - Sort order is: Sauce Labs Backpack, Sauce Labs Bike Light, Sauce Labs Bolt T-Shirt, Sauce Labs Fleece Jacket, Sauce Labs Onesie, Test.allTheThings() T-Shirt (Red)
 - Sort selection is clearly shown
- **Post-conditions:** Products are sorted by name A-Z

TC-016: Sort Products by Name (Z to A)

- **User Story:** US-2.1.3
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is logged in and on inventory page
- **Test Steps:**
 - Click on sort dropdown
 - Select "Name (Z to A)"
 - Check product order
- **Expected Result:**
 - Products are sorted alphabetically by name in reverse order
 - Sort order is reverse of A-Z sorting
 - Sort selection is clearly shown
- **Post-conditions:** Products are sorted by name Z-A

TC-017: Sort Products by Price (Low to High)

- **User Story:** US-2.1.3
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is logged in and on inventory page
- **Test Steps:**
 - Click on sort dropdown
 - Select "Price (low to high)"
 - Check product order by price
- **Expected Result:**
 - Products are sorted by price from lowest to highest
 - Price order is maintained correctly
 - Sort selection is clearly shown
- **Post-conditions:** Products are sorted by price low-high

TC-018: Sort Products by Price (High to Low)

- **User Story:** US-2.1.3
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is logged in and on inventory page
- **Test Steps:**
 - Click on sort dropdown
 - Select "Price (high to low)"
 - Check product order by price
- **Expected Result:**
 - Products are sorted by price from highest to lowest
 - Price order is maintained correctly
 - Sort selection is clearly shown
- **Post-conditions:** Products are sorted by price high-low

TC-019: Product Information Consistency

- **User Story:** US-2.1.4
- **Priority:** High

- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is logged in and on inventory page
- **Test Steps:**
 - Note product details on inventory page for "Sauce Labs Backpack"
 - Click on product name to view details
 - Compare information between inventory and detail pages
 - Return to inventory and check consistency
- **Expected Result:**
 - Product name matches between pages
 - Product price matches between pages
 - Product description matches between pages
 - Product image matches between pages
- **Post-conditions:** Product information consistency is verified

TC-020: Navigation Menu Access

- **User Story:** US-2.2.1
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is logged in and on inventory page
- **Test Steps:**
 - Click hamburger menu (three lines icon)
 - Check that menu options are displayed
 - Check that all menu items are clickable
- **Expected Result:**
 - Menu slides out from left side
 - Menu options displayed: All Items, About, Logout, Reset App State
 - All menu items are clickable and work
 - Menu can be closed by clicking X or outside area
- **Post-conditions:** Navigation menu is accessible

TC-021: Back to Products Navigation

- **User Story:** US-2.2.2
- **Priority:** Medium
- **Severity:** Medium

- **Test Type:** Functional
- **Preconditions:** User is on product detail page
- **Test Steps:**
 - From product detail page, click "Back to products"
 - Check return to inventory page
 - Check if previous sort order is kept
- **Expected Result:**
 - User is returned to inventory page
 - All products are displayed
 - Previous sort order is kept (if any)
 - Navigation works smoothly
- **Post-conditions:** User is back on inventory page

TC-022: Product Images Load Correctly

- **User Story:** US-2.1.1
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is logged in and on inventory page
- **Test Steps:**
 - Check that all product images are loaded
 - Check image quality and visibility
 - Check that no broken image placeholders exist
- **Expected Result:**
 - All 6 product images are loaded successfully
 - Images are clear and properly sized
 - No broken image icons or placeholders
 - Images match product names
- **Post-conditions:** All product images are properly displayed

3.3 Shopping Cart Management (15 Test Cases)

TC-023: Add Single Product to Cart from Inventory

- **User Story:** US-3.1.1
- **Priority:** Critical

- **Severity:** High
- **Test Type:** Functional
- **Preconditions:** User is logged in and on inventory page
- **Test Steps:**
 - Click "Add to cart" button for "Sauce Labs Backpack"
 - Check that cart badge updates
 - Check that button text changes
- **Expected Result:**
 - Cart badge shows "1"
 - Button text changes to "Remove"
 - Product is added to cart
 - Visual feedback is provided
- **Post-conditions:** One product is in cart

TC-024: Add Single Product to Cart from Product Detail

- **User Story:** US-3.1.1
- **Priority:** High
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is on product detail page
- **Test Steps:**
 - Click "Add to cart" button on product detail page
 - Check that cart badge updates
 - Check that button text changes
- **Expected Result:**
 - Cart badge shows "1"
 - Button text changes to "Remove"
 - Product is added to cart
 - User can continue shopping or go to cart
- **Post-conditions:** One product is in cart

TC-025: Add Multiple Products to Cart

- **User Story:** US-3.1.2
- **Priority:** High
- **Severity:** Medium
- **Test Type:** Functional

- **Preconditions:** User is logged in and on inventory page
- **Test Steps:**
 - Click "Add to cart" for "Sauce Labs Backpack"
 - Click "Add to cart" for "Sauce Labs Bike Light"
 - Click "Add to cart" for "Sauce Labs Bolt T-Shirt"
 - Check that cart badge updates
- **Expected Result:**
 - Cart badge shows "3"
 - Each product button changes to "Remove"
 - All products are added to cart
 - Cart counter increases with each addition
- **Post-conditions:** Three products are in cart

TC-026: Remove Product from Cart via Inventory

- **User Story:** US-3.2.2
- **Priority:** High
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** Product is added to cart from inventory page
- **Test Steps:**
 - Click "Remove" button for added product
 - Check that cart badge updates
 - Check that button text changes back
- **Expected Result:**
 - Cart badge decreases by 1
 - Button text changes back to "Add to cart"
 - Product is removed from cart
 - Cart counter updates immediately
- **Post-conditions:** Product is removed from cart

TC-027: View Cart Contents

- **User Story:** US-3.2.1
- **Priority:** Critical
- **Severity:** High
- **Test Type:** Functional
- **Preconditions:** At least one product is added to cart

- **Test Steps:**
 - Click shopping cart icon
 - Check that cart page is displayed
 - Check that product details are shown
 - Check quantity and pricing
- **Expected Result:**
 - Cart page is displayed
 - All added products are listed
 - Product name, description, and price are shown
 - Quantity is displayed (1 per item)
 - "Remove" button is available for each item
- **Post-conditions:** Cart contents are visible

TC-028: Remove Product from Cart Page

- **User Story:** US-3.2.2
- **Priority:** High
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is on cart page with products
- **Test Steps:**
 - Click "Remove" button for a product
 - Check that product is removed from cart
 - Check that cart badge updates
 - Check that total updates
- **Expected Result:**
 - Product is removed from cart page
 - Cart badge decreases by 1
 - Remaining products are still displayed
 - Page updates immediately
- **Post-conditions:** Product is removed from cart

TC-029: Continue Shopping from Cart

- **User Story:** US-3.3.2
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Functional

- **Preconditions:** User is on cart page
- **Test Steps:**
 - Click "Continue Shopping" button
 - Check return to inventory page
 - Check that cart contents are saved
- **Expected Result:**
 - User is moved to inventory page
 - Cart badge shows same count
 - Previously added products stay in cart
 - User can continue adding products
- **Post-conditions:** User is back on inventory page with cart saved

TC-030: Cart Persistence Across Pages

- **User Story:** US-3.2.3
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User has added products to cart
- **Test Steps:**
 - Add product to cart from inventory
 - Go to product detail page
 - Go back to inventory
 - Check cart badge and contents
- **Expected Result:**
 - Cart badge shows consistent count
 - Cart contents are saved
 - Added products stay in cart
 - No items are lost during navigation
- **Post-conditions:** Cart contents are maintained

TC-031: Empty Cart Display

- **User Story:** US-3.2.4
- **Priority:** Low
- **Severity:** Low
- **Test Type:** Functional
- **Preconditions:** User is on cart page with no products

- **Test Steps:**
 - Go to cart page when empty
 - Check empty cart message or state
 - Check continue shopping option
- **Expected Result:**
 - Cart page shows empty state
 - Continue shopping option is available
 - Cart badge shows no count or "0"
 - Checkout button is not available or disabled
- **Post-conditions:** Empty cart state is displayed

TC-032: Cart Access from Any Page

- **User Story:** US-3.3.1
- **Priority:** High
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is logged in and has products in cart
- **Test Steps:**
 - From inventory page, click cart icon
 - Check that cart page is accessible
 - From product detail page, click cart icon
 - Check that cart page is accessible
- **Expected Result:**
 - Cart icon is visible on all pages
 - Cart is accessible with one click
 - Cart badge shows current item count
 - Navigation to cart is consistent
- **Post-conditions:** Cart is accessible from any page

TC-033: Multiple Same Product Addition

- **User Story:** US-3.1.3
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is logged in and on inventory page
- **Test Steps:**

- Click "Add to cart" for "Sauce Labs Backpack"
- Check that button changes to "Remove"
- Try to add same product again
- Check behavior
- **Expected Result:**
 - First click adds product to cart
 - Button changes to "Remove"
 - Second click removes product from cart
 - Cart badge updates correctly
- **Post-conditions:** Product addition/removal works correctly

TC-034: Cart Badge Accuracy

- **User Story:** US-3.1.2
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is logged in
- **Test Steps:**
 - Start with empty cart
 - Add products one by one
 - Check that cart badge updates with each addition
 - Remove products and check that badge updates
- **Expected Result:**
 - Cart badge starts at 0 or no display
 - Badge increases with each product addition
 - Badge decreases with each product removal
 - Badge count is always accurate
- **Post-conditions:** Cart badge accurately shows cart contents

TC-035: Cart State After Logout/Login

- **User Story:** US-3.2.3
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User has products in cart
- **Test Steps:**

- Add products to cart
- Logout from application
- Login again with same user
- Check cart state
- **Expected Result:**
 - Cart is cleared after logout
 - Fresh session starts with empty cart
 - Cart badge shows 0 or no display
 - Previous cart contents are not restored
- **Post-conditions:** Cart is reset after new login

TC-036: Add All Products to Cart

- **User Story:** US-3.1.2
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is logged in and on inventory page
- **Test Steps:**
 - Add all 6 products to cart
 - Check that cart badge updates to 6
 - Go to cart page
 - Check that all products are listed
- **Expected Result:**
 - All 6 products are added to cart
 - Cart badge shows "6"
 - Cart page displays all 6 products
 - Each product has remove button
- **Post-conditions:** All products are in cart

TC-037: Remove All Products from Cart

- **User Story:** US-3.2.2
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** All products are in cart
- **Test Steps:**

- From cart page, remove all products one by one
- Check that cart badge updates with each removal
- Check empty cart state
- **Expected Result:**
 - Each product is removed successfully
 - Cart badge decreases with each removal
 - Final state shows empty cart
 - Continue shopping option is available
- **Post-conditions:** Cart is empty

3.4 Checkout Process (10 Test Cases)

TC-038: Checkout Process Start

- **User Story:** US-4.1.1
- **Priority:** Critical
- **Severity:** High
- **Test Type:** Functional
- **Preconditions:** User has products in cart
- **Test Steps:**
 - Go to cart page
 - Click "Checkout" button
 - Check that checkout information page is displayed
- **Expected Result:**
 - Checkout information page is displayed
 - Form fields are present: First Name, Last Name, Postal Code
 - Cancel and Continue buttons are available
 - URL changes to /checkout-step-one
- **Post-conditions:** User is on checkout information page

TC-039: Complete Checkout with Valid Information

- **User Story:** US-4.1.1
- **Priority:** Critical
- **Severity:** High
- **Test Type:** Functional
- **Preconditions:** User is on checkout information page

- **Test Steps:**
 - Enter "John" in First Name field
 - Enter "Doe" in Last Name field
 - Enter "12345" in Postal Code field
 - Click "Continue" button
- **Expected Result:**
 - Form is submitted successfully
 - User is moved to checkout overview page
 - Order summary is displayed
 - URL changes to /checkout-step-two
- **Post-conditions:** User is on checkout overview page

TC-040: Checkout with Missing First Name

- **User Story:** US-4.1.2
- **Priority:** High
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is on checkout information page
- **Test Steps:**
 - Leave First Name field empty
 - Enter "Doe" in Last Name field
 - Enter "12345" in Postal Code field
 - Click "Continue" button
- **Expected Result:**
 - Error message displayed: "Error: First Name is required"
 - User stays on checkout information page
 - Form data is saved
 - Error message is clearly visible
- **Post-conditions:** User stays on checkout form with error

TC-041: Checkout with Missing Last Name

- **User Story:** US-4.1.2
- **Priority:** High
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is on checkout information page

- **Test Steps:**
 - Enter "John" in First Name field
 - Leave Last Name field empty
 - Enter "12345" in Postal Code field
 - Click "Continue" button
- **Expected Result:**
 - Error message displayed: "Error: Last Name is required"
 - User stays on checkout information page
 - Form data is saved
 - Error message is clearly visible
- **Post-conditions:** User stays on checkout form with error

TC-042: Checkout with Missing Postal Code

- **User Story:** US-4.1.2
- **Priority:** High
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is on checkout information page
- **Test Steps:**
 - Enter "John" in First Name field
 - Enter "Doe" in Last Name field
 - Leave Postal Code field empty
 - Click "Continue" button
- **Expected Result:**
 - Error message displayed: "Error: Postal Code is required"
 - User stays on checkout information page
 - Form data is saved
 - Error message is clearly visible
- **Post-conditions:** User stays on checkout form with error

TC-043: Checkout Overview Display

- **User Story:** US-4.2.1
- **Priority:** Critical
- **Severity:** High
- **Test Type:** Functional
- **Preconditions:** User has completed checkout information

- **Test Steps:**
 - Check that checkout overview page is displayed
 - Check that all cart items are listed
 - Check pricing information
 - Check total calculation
- **Expected Result:**
 - All cart items are displayed with details
 - Item total (subtotal) is calculated correctly
 - Tax amount is calculated and displayed
 - Total amount is sum of subtotal and tax
 - Finish and Cancel buttons are available
- **Post-conditions:** Order summary is displayed for review

TC-044: Complete Purchase

- **User Story:** US-4.3.1
- **Priority:** Critical
- **Severity:** High
- **Test Type:** Functional
- **Preconditions:** User is on checkout overview page
- **Test Steps:**
 - Review order details
 - Click "Finish" button
 - Check order completion page
- **Expected Result:**
 - Order completion page is displayed
 - Success message is shown
 - Order confirmation details are visible
 - Thank you message is displayed
- **Post-conditions:** Order is completed successfully

TC-045: Cancel Checkout from Information Page

- **User Story:** US-4.1.3
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is on checkout information page

- **Test Steps:**
 - Click "Cancel" button
 - Check return to cart page
 - Check that cart contents are saved
- **Expected Result:**
 - User is moved to cart page
 - Cart contents are saved
 - Checkout process is cancelled
 - User can continue shopping or retry checkout
- **Post-conditions:** User is back on cart page

TC-046: Cancel Checkout from Overview Page

- **User Story:** US-4.2.2
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is on checkout overview page
- **Test Steps:**
 - Click "Cancel" button
 - Check return to cart page
 - Check that cart contents are saved
- **Expected Result:**
 - User is moved to cart page
 - Cart contents are saved
 - Checkout process is cancelled
 - User can continue shopping or retry checkout
- **Post-conditions:** User is back on cart page

TC-047: Order Total Calculation Accuracy

- **User Story:** US-4.2.3
- **Priority:** High
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User is on checkout overview page with multiple items
- **Test Steps:**
 - Add items with different prices to cart
 - Go to checkout overview
 - Check subtotal calculation
 - Check tax calculation
 - Check total calculation
- **Expected Result:**
 - Subtotal equals sum of all item prices
 - Tax is calculated as 8% of subtotal
 - Total equals subtotal plus tax
 - All calculations are accurate to 2 decimal places
- **Post-conditions:** Order calculations are verified

3.5 User Experience and Interface (3 Test Cases)

TC-048: Cross-Browser Compatibility - Chrome

- **User Story:** US-5.1.2
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Non-functional
- **Preconditions:** Chrome browser is available and updated
- **Test Steps:**
 - Open SauceDemo in Chrome browser
 - Do complete user journey (login → browse → add to cart → checkout)
 - Check that all functions work correctly
 - Check visual appearance and layout
- **Expected Result:**
 - All functions work as expected
 - Visual layout is correct and consistent
 - No browser-specific errors occur
 - Performance is acceptable
- **Post-conditions:** Chrome compatibility is verified

TC-049: Cross-Browser Compatibility - Firefox

- **User Story:** US-5.1.2
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Non-functional
- **Preconditions:** Firefox browser is available and updated
- **Test Steps:**
 - Open SauceDemo in Firefox browser
 - Do complete user journey (login → browse → add to cart → checkout)
 - Check that all functions work correctly
 - Check visual appearance and layout
- **Expected Result:**
 - All functions work as expected
 - Visual layout is correct and consistent

- No browser-specific errors occur
- Performance is acceptable
- **Post-conditions:** Firefox compatibility is verified

TC-050: Error Message Display and Dismissal

- **User Story:** US-5.2.1
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Functional
- **Preconditions:** User can create error conditions
- **Test Steps:**
 - Create login error with wrong credentials
 - Check that error message is displayed clearly
 - Create checkout validation error
 - Check that error message is displayed clearly
 - Test error message dismissal (X button)
- **Expected Result:**
 - Error messages are clear and user-friendly
 - Error messages give helpful information
 - Error messages can be dismissed
 - Error styling is consistent across the application
- **Post-conditions:** Error handling is verified

3.6 Security and Performance (2 Test Cases)

TC-051: Basic Input Validation

- **User Story:** US-6.1.1
- **Priority:** High
- **Severity:** High
- **Test Type:** Security
- **Preconditions:** User has access to login and checkout forms
- **Test Steps:**
 - Test login form with special characters in username
 - Test login form with SQL injection patterns
 - Test checkout form with script tags in input fields

- Test form with very long input strings
- **Expected Result:**
 - Special characters are handled properly
 - SQL injection attempts are prevented
 - Script tags are cleaned or rejected
 - Input length limits are enforced
 - No security problems are exposed
- **Post-conditions:** Basic input validation is verified

TC-052: Page Load Performance

- **User Story:** US-6.2.1
- **Priority:** Medium
- **Severity:** Medium
- **Test Type:** Performance
- **Preconditions:** Network conditions are stable
- **Test Steps:**
 - Measure login page load time
 - Measure inventory page load time after login
 - Measure product detail page load time
 - Measure cart page load time
 - Measure checkout pages load time
- **Expected Result:**
 - Login page loads within 3 seconds
 - Inventory page loads within 3 seconds
 - Product detail page loads within 2 seconds
 - Cart page loads within 2 seconds
 - Checkout pages load within 3 seconds
- **Post-conditions:** Performance benchmarks are established

4. Test Execution Summary

4.1 Test Case Distribution

Function Area	Test Cases	Critical	High	Medium	Low
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Login & Access	12	1	6	4	1
Product Catalog	10	1	2	6	1
Shopping Cart	15	2	4	8	1
Checkout Process	10	4	4	2	0
User Experience	3	0	0	3	0
Security & Performance	2	0	1	1	0
TOTAL	52	8	17	24	3

4.2 User Story Coverage

Function Group	User Stories	Test Cases	Coverage
User Login	6	12	200%
Product Catalog	6	10	167%
Shopping Cart	8	15	188%
Checkout Process	9	10	111%
User Experience	6	3	50%
Security & Performance	4	2	50%

4.3 Test Types Distribution

- **Functional Testing:** 47 test cases (90%)
- **Non-functional Testing:** 3 test cases (6%)
- **Security Testing:** 1 test case (2%)
- **Performance Testing:** 1 test case (2%)

4.4 Acceptance Criteria Coverage

Each test case is directly connected to specific acceptance criteria from the user stories. This ensures:

- **100% Requirement Coverage:** All acceptance criteria are tested

- **Positive and Negative Scenarios:** Both normal behavior and error conditions
- **Edge Cases:** Boundary conditions and unusual inputs
- **Integration Scenarios:** Complete workflow validation

5. Test Execution Guidelines

5.1 Execution Order

Phase 1: Critical Path (Priority: Critical)

- Execute test cases TC-001, TC-013, TC-023, TC-027, TC-038, TC-039, TC-043, TC-044

Phase 2: Core Functions (Priority: High)

- Execute all remaining High priority test cases
- Focus on login, cart management, and checkout

Phase 3: Extended Testing (Priority: Medium)

- Execute Medium priority test cases
- Include cross-browser and edge case testing

Phase 4: Additional Validation (Priority: Low)

- Execute Low priority test cases
- Do final validation and cleanup testing

5.2 Environment Requirements

- **Browsers:** Chrome (latest), Firefox (latest), Edge (latest)
- **Test Data:** Standard user accounts provided by SauceDemo
- **Network:** Stable internet connection for consistent performance testing
- **Tools:** Test automation framework, performance monitoring tools

5.3 Entry Criteria

- SauceDemo website is accessible and stable
- Test environment is configured and validated
- Test data is prepared and available

- Testing tools are installed and configured

5.4 Exit Criteria

- All Critical and High priority test cases pass (100%)
- Medium priority test cases pass rate $\geq 95\%$
- All found defects are documented and categorized
- Performance benchmarks are established
- Cross-browser compatibility is verified

6. Defect Management

6.1 Defect Reporting Template

Defect ID: DEF-XXX

Test Case ID: TC-XXX

Severity: Critical/High/Medium/Low

Priority: P1/P2/P3/P4

Environment: Browser, OS, Version

Summary: Brief description

Steps to Reproduce: Detailed steps

Expected Result: What should happen

Actual Result: What actually happens

Attachments: Screenshots, logs

6.2 Defect Tracking

- **Critical/High Severity:** Must be fixed before release
- **Medium Severity:** Should be fixed in current iteration
- **Low Severity:** Can be fixed in future iterations
- **Blocker:** Prevents further testing, highest priority

7. Requirements to Test Cases Connection

7.1 Requirement to Test Case Mapping

User Story	Test Cases	Coverage Status
US-1.1.1	TC-001, TC-010, TC-011	☒ Complete
US-1.1.2	TC-002, TC-003, TC-012	☒ Complete
US-1.1.3	TC-004	☒ Complete
US-1.1.4	TC-005, TC-006, TC-007	☒ Complete
US-1.2.1	TC-009	☒ Complete
US-1.2.2	TC-008, TC-035	☒ Complete
US-2.1.1	TC-013, TC-022	☒ Complete
US-2.1.2	TC-014	☒ Complete
US-2.1.3	TC-015, TC-016, TC-017, TC-018	☒ Complete
US-2.1.4	TC-019	☒ Complete
US-2.2.1	TC-020	☒ Complete
US-2.2.2	TC-021	☒ Complete
US-3.1.1	TC-023, TC-024	☒ Complete
US-3.1.2	TC-025, TC-034, TC-036	☒ Complete
US-3.1.3	TC-033	☒ Complete
US-3.2.1	TC-027	☒ Complete
US-3.2.2	TC-026, TC-028, TC-037	☒ Complete
US-3.2.3	TC-030, TC-035	☒ Complete
US-3.2.4	TC-031	☒ Complete

US-3.3.1	TC-032	☒ Complete
US-3.3.2	TC-029	☒ Complete
US-4.1.1	TC-038, TC-039	☒ Complete
US-4.1.2	TC-040, TC-041, TC-042	☒ Complete
US-4.1.3	TC-045	☒ Complete
US-4.2.1	TC-043	☒ Complete
US-4.2.2	TC-046	☒ Complete
US-4.2.3	TC-047	☒ Complete
US-4.3.1	TC-044	☒ Complete
US-5.1.2	TC-048, TC-049	☒ Complete
US-5.2.1	TC-050	☒ Complete
US-6.1.1	TC-051	☒ Complete
US-6.2.1	TC-052	☒ Complete

7.2 Test Coverage Summary

- **Total User Stories:** 39
- **Covered User Stories:** 32 (82%)
- **Total Test Cases:** 52
- **Critical Test Cases:** 8 (15%)
- **High Priority Test Cases:** 17 (33%)
- **Medium Priority Test Cases:** 24 (46%)
- **Low Priority Test Cases:** 3 (6%)

8. Next Steps

1. ☒ **Test Strategy** - Complete
2. ☒ **User Story Mapping** - Complete
3. ☒ **Test Case Development** - Complete (52 test cases)
4. ☐ **Next: Framework Architecture Design** - Technical implementation plan

5. **Pending: Setup and Installation Guide** - Environment configuration
6. **Pending: Test Automation Implementation** - Actual framework development